INCH-POUND

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MILITARY SPECIFICATION

THRUSTER, CARTRIDGE ACTUATED, M16 REMANUFACTURED METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M16 Cartridge Actuated Thruster Remanufactured Metal Parts Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-413	Visual Inspection Guide for Elastomeric O-rings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

Naval Surface Warfare Center, Indian Head (CAGE Codes 19200 and 19203)

10521322	Thruster, Cartridge Actuated, M16, Metal Parts Assembly
8845243	Box, Packing, Ammunition, For Thruster, Cartridge Actuated, M16
8845244	Box, Fiberboard (Inner), Packing, Ammunition, For Thruster, Cartridge Actuated, M16
9217557	Box, Fiberboard (Outer), Alternative, Packing Ammunition, for Thruster, Cartridge Actuated, M16

U.S. Air Force

TPO 00-849-0214	Thruster, Cartridge Actuated, M16

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Data Control Branch (Code 8410), Indian Head, MD 20640-5035.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the thrusters shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal. The remanufactured thruster metal parts assembly shall contain the following new parts or items: shear pin, O-rings, quad rings, setscrews, spring, silicone fluid, nylon pellets, and lock nut.

3.2 Hydrostatic pressure. The cylinder and head shall withstand a hydrostatic pressure of 10,000 pounds per square inch (psi) for 15 seconds, minimum, without leaking, deformation, or mechanical failure (see 6.3.1).

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3.3 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of configuration. Remanufactured exterior components (see 6.3.2) shall have the existing protective finish stripped and the specified protective finish reapplied. When there are only small surface areas of exposed base metal on the aluminum exterior components (body assembly and end cap) it may be permissible to treat the surfaces with chemical film Class 1A as specified in MIL-C-5541.

3.4 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction using the procedures of 4.5.5.

3.5 Shear pin strength. The shear pin, after application of protective finish, shall shear when a load of 41 $\pm$ 5 pounds is applied at the steady rate of 0.125 inch per minute.

3.6 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing.

3.6.1 Integrated plant products. When remanufacturing and loading of the assembly is performed in an integrated plant, the metal parts assembly requirement may be waived and the residual magnetism requirement performed on the loaded assembly only.

3.6.2 Threaded component parts. Class 2A or 2B thread gage shall be used on remanufactured component parts inspection versus class 3A or 3B thread gage required on new hardware for thrusters.

3.6.3 Remanufactured components. Remanufactured components which require modification shall be modified in accordance with instructions noted in 6.5.

3.6.4 Oil leakage. Prior to assembly of the head and cartridge, a load of 20 $\pm$ 2 pounds, for 15 seconds minimum, shall be applied to the floating piston to determine presence of O-ring and quad ring. There shall be no evidence of oil leakage.

3.6.5 Head and cylinder. The head and cylinder shall be assembled with sufficient torque to secure the head.

3.6.6 Presence of piston, shear pin. Upon completion of assembly, a load of 20 $\pm$ 2 pounds shall be applied to the locking pin to determine if the shear pin is properly engaged.

3.7 Workmanship. Thrusters shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.7.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.7.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

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3.7.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: room ambient (68 to 77°F)
b. Altitude: sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric pressure: 14.7 psi

4.3 Quality conformance inspection.

4.3.1 Lot.

4.3.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with MIL-STD-105.

4.3.1.2 Size of lot. For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 assemblies, maximum, provided the homogeneity requirements of inspection lots have been retained.

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4.3.2 Examination. On reprocessed parts, one hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4x) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.3.2.1 Examination of reclaimed parts. Examination of reusable parts, which require no reprocessing, shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be reprocessed or rejected.

4.3.2.2 Pin, Firing, Drawing 8596416, covering a detail of Drawing 10521322.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Radius	Gage
102	Metal defective	Visual
103	Protective finish inadequate or defective	Visual
104	Burr or sharp corner	Visual
105	Foreign matter	Visual
Minor		
None defined		

4.3.2.3 Piston, Floating, Drawing 8596417, covering a detail of Drawing 10521322.

Critical		
None defined		
Major		
101	Pitch diameter, max	Gage
102	Thread damaged or not full	Visual
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Burr or sharp corner	Visual
106	Foreign matter	Visual

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4.3.2.4 Piston, Drawing 8596418, covering a detail of Drawing 10521322.

Critical
None defined

Major		
101	Pitch diameter, max	Gage
102	Thread damaged or not full	Visual
103	Metal defective	Visual
104	Protective finish improper	Visual
105	Burr or sharp corner	Visual
106	Foreign matter	Visual

Minor
None defined

4.3.2.5 Clevis, Drawing 8596421, covering a detail of Drawing 10521322.

Critical
None defined

Major		
101	Pitch diameter, max	Gage
102	Thread damaged or not full	Visual
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Burr or sharp corner	Visual
106	Foreign matter	Visual

Minor
None defined

4.3.2.6 Pin, Locking, Drawing 8596422, covering a detail of Drawing 10521322.

Critical
None defined

Major		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Burr or sharp corner	Visual
104	Foreign matter	Visual

Minor
None defined

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4.3.2.7 Head, Drawing 10521324, covering a detail of Drawing 10521322.

Critical		
1	Surface finish of long cavity improper	Gage
Major		
101	Pitch diameter of large thread, max	Gage
102	Pitch diameter of inlet port thread, max	Gage
103	Pitch diameter of setscrew thread, max	Gage
104	Thread damaged or not full	Visual
105	Metal defective	Visual
106	Protective finish inadequate or defective	Visual
107	Burr or sharp edges	Visual
108	Foreign matter	Visual
Minor		
None defined		

4.3.2.8 Cartridge Retainer, Drawing 10521325, covering a detail of Drawing 10521322.

Critical		
None defined		
Major		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Foreign matter	Visual
104	Burr or sharp corner	Visual
Minor		
None defined		

4.3.2.9 Cylinder and Pellet Assembly, Drawing 10521323, covering a detail of Drawing 10521322.

Critical		
None defined		
Major		
101	Pitch diameter of large thread, max	Gage
102	Pitch diameter of setscrew thread, max	Gage
103	Protrusion of pellets, max	Gage
104	Pellet loose, missing, or improperly inserted	Visual
105	Thread damaged	Visual
106	Foreign matter	Visual
107	Burr or sharp corners	Visual

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4.3.2.10 Pin, Shear, Drawing 8596419, covering a detail of Drawing 10521322.

Critical
None defined

Major		
101	Diameter, max	Gage
102	Total length, min	Gage
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Burr or sharp corner	Visual
106	Foreign matter	Visual
Minor		
201	Total length, max	Gage
202	Surface finish improper	Visual

4.3.2.11 Pin, Shear, Drawing 8596423, covering a detail of Drawing 10521322.

Critical
None defined

Major		
101	Diameter, max	Gage
102	Total length, min	Gage
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Burr or sharp corner	Visual
106	Foreign matter	Visual
Minor		
201	Total length, max	Gage
202	Surface finish improper	Visual

4.3.2.12 Spring, Drawing 8596424, covering a detail of Drawing 10521322.

Critical
None defined

Major		
101	Load at short compressed length	Gage
102	Load at long compressed length	Gage
103	Solid length, max	Gage
104	Ends incorrect	Visual
105	Protective finish inadequate or defective	Visual
106	Burr, sharp edges	Visual

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107	Foreign matter	Visual
Minor		
201	Outside diameter	Gage
202	Fit free into hole	Gage
203	Fit free over rod	Gage

4.3.2.13 Assembly, Drawing 10521322.

Critical
None defined

Major		
101	Any part missing, except as otherwise classified	Visual
102	Improper assembly	Visual
103	Evidence of oil leakage	Visual
104	Metal defective	Visual
105	Burr	Visual
106	Foreign matter	Visual
Minor		
201	Protective plug missing	Visual
202	Protective finish inadequate or defective	Visual

4.3.2.14 Box, Fiberboard (Inner), Packing, Ammunition, Drawing 8845244.

Critical
None defined

Major
None defined

Minor		
201	Contents loose	Manual
202	Box damaged	Visual
203	Marking missing, incorrect, or illegible	Visual
204	Cut, damaged, or improperly sealed bag	Visual

4.3.2.15 Box, Packing, Ammunition, Drawing 8845243.

Critical
None defined

Major
None defined

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Minor		
201	Contents loose	Visual/Manual
202	Strapping missing, loose, broken, or improperly assembled	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.3.2.16 Box, Fiberboard (Outer), Alternative, Packing, Ammunition, Drawing 9217557.

Critical
None defined

Major
None defined

Minor		
201	Contents loose	Manual/Visual
202	Hardware missing, loose, broken, or improperly engaged	Visual
203	Strapping missing, loosed, broken, or improperly engaged	Visual
204	Contents of box exposed	Visual
205	Box damaged	Visual
206	Marking missing, incorrect, or illegible	Visual

4.4 Inspection testing. The following tests shall be performed in accordance with 4.5.

4.4.1 Hydrostatic pressure. Each cylinder and head shall be subjected to the hydrostatic pressure test. Failure of any unit to comply with 3.2 shall be cause for rejection of the component.

4.4.2 Oil leakage. Each subassembly shall be subjected to an oil leakage test. Failure of any unit to comply with the requirement of 3.6.4 shall be cause for rejection of the unit.

4.4.3 Presence of piston shear pin. Each assembly shall be subjected to a test to determine the presence of a shear pin in the piston lock. Failure of any unit to comply with the requirement of 3.6.6 shall be cause for rejection of the unit.

4.4.4 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.4.5 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism as specified in 4.5.5. Failure of any unit in the inspection sample to comply with 3.4 shall be cause for rejection of the assembly.

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4.4.6 Shear pin strength. Fifty pins from each lot of shear pins shall be tested for shear strength. Failure of any shear pin to comply with 3.5 shall be cause for rejection of the lot.

4.5 Test methods and procedures.

4.5.1 Hydrostatic pressure. Prior to application of protective finish the cylinder and head shall be mounted in the device, using a fixture for properly sealing all openings, and subjected to the internal hydrostatic pressure to determine compliance with 3.2.

4.5.2 Oil leakage. Prior to assembly of the head and cartridge retainer, the subassembly shall be secured in a device and a load of 20 ± 2 pounds, for 15 seconds minimum, shall be applied to the floating piston.

4.5.3 Presence of piston shear pin. Upon completion of assembly, the thruster shall be secured in a device and a load of 20 ± 2 pounds shall be applied to the locking pin to determine that the shear pin is properly engaged in the pin lock.

4.5.4 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.5.5 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (head end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.5.6 Shear pin strength. After application of protective finish, the pin shall be subjected to the shear pin strength test to determine compliance with 3.5. The test shall be performed in a fixture in conjunction with a tensile testing machine.

4.6 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be Level A or B, as specified by the contracting activity (see 6.2). Level A shall be in accordance with Drawings 8845243 and 8845244. Level B shall be in accordance with Drawings 8845244 and 9217557.

5.2 Air Force packaging. Air Force packaging shall be in accordance with TPO 00-849-0214.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The thruster assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

c. The responsible inspection activity (see 4.1)

d. Any special instruction requirements (see 4.2)

e. Level of packing and marking (see 5.1)

6.3 Definitions.

6.3.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3.2 Exterior components. Remanufactured exterior components are: head, cylinder, clevis, and locking pin.

6.4 Old engraved or stamped lot and serial numbers. Old serial numbers and lot numbers engraved or stamped on metal parts shall be over-stamped by x's to obliterate those markings before applications of final protective finish.

6.5 Special instructions. Special instructions covering the modification, repair, and assembly of remanufactured components shall be as follows:

a. Bodies which contain nylon pellets and are to be replated shall have the pellets removed and new pellets installed after replating in accordance with the applicable drawing.

b. Head components with straight through firing pin bore shall be replaced with Head, Part No. 10521324.

c. Cylinder components with .945 $\pm$ .002 diameter shoulder at the thread end shall be replace with Cylinder, Part No. 10521323 (1.058 $\pm$ .002 diameter shoulder).

d. All spacer components shall be replace with Cartridge Retainer, Part No. 10521325.

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6.6 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.7 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:
Navy - OS
Air Force - 99

Review activity:
Air Force - 70

Preparing activity
Navy - OS
(Project 1377-0E24)

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INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-60680A	93/04/12

3. DOCUMENT TITLE

THRUSTER, CARTRIDGE ACTUATED, M16 REMANUFACTURED METAL PARTS ASSEMBLY

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Commander, Indian Head Division Naval Surface Warfare Center	(1) Commercial (301) 743-4358/4510	(2) AUTOVON 354-4358/4510
c. ADDRESS (Include Zip Code) Code 8420 101 Strauss Avenue Indian Head, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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